

Prescribing the ice–regolith contact angle

Wall free energy, its weak form, and why no surface energy survives into the residual coefficient

lunar_regolith_DSM

2026-09-16

Summary. The substrate in this two-phase model is the domain wall, not a phase field, so it must act through a boundary condition. Cahn’s wall free energy $f_w(\phi) = \sigma_{as} + (\sigma_{is} - \sigma_{as})h(\phi)$ joins the bulk functional as an ordinary surface term — both are energies, both in $\text{J}/\text{m}^2 \times \text{m}$, and they add directly. Its variation supplies a natural condition on $\partial_n \phi$ that fixes the contact angle. The resulting residual term is $-3M \cos \theta h'(\phi)/6$, with no ϵ and no surface energy in the coefficient. Both cancellations are derived below, and neither requires a unit conversion.

1. The free energy

Let ϕ be the ice fraction, $\phi = 1$ ice and $\phi = 0$ vapour, and let $\Gamma \subseteq \partial\Omega$ be the part of the boundary representing regolith. Write the whole free energy physically, in energy units throughout:

$$\mathcal{F}[\phi] = \underbrace{\int_{\Omega} \left[3\sigma_{ia}\epsilon |\nabla\phi|^2 + \frac{3\sigma_{ia}}{\epsilon} \phi^2(1-\phi)^2 \right] d\Omega}_{\mathcal{F}_{\text{bulk}}} + \underbrace{\oint_{\Gamma} f_w(\phi) d\Gamma}_{\mathcal{F}_{\text{wall}}}, \quad (1)$$

$$f_w(\phi) = \sigma_{as}(1 - h(\phi)) + \sigma_{is}h(\phi) = \sigma_{as} + (\sigma_{is} - \sigma_{as})h(\phi), \quad h(\phi) = \phi^2(3 - 2\phi). \quad (2)$$

Both terms of (1) are energies — in 2D, J per unit depth — so they add with no conversion between them. This is worth stating because it is easy to reach for one: the form of $\mathcal{F}_{\text{bulk}}$ implemented in the code (§3) is non-dimensionalised and is *not* an energy, and mixing that form with a physical f_w does require a conversion. Writing both physically, as here, avoids the question entirely.

The bulk coefficients in (1) are fixed by requiring that the ice–vapour interface carry surface energy σ_{ia} . For the equilibrium profile

$$\phi(x) = \frac{1}{2} \left(1 + \tanh \frac{x}{2\epsilon} \right), \quad \phi' = \frac{\phi(1-\phi)}{\epsilon}, \quad (3)$$

the two bulk terms are equal (equipartition, from the first integral of the Euler–Lagrange equation), and

$$\int_{-\infty}^{\infty} \left[3\sigma_{ia}\epsilon \phi'^2 + \frac{3\sigma_{ia}}{\epsilon} \phi^2(1-\phi)^2 \right] dx = 6\sigma_{ia}\epsilon \int \phi'^2 dx = 6\sigma_{ia}\epsilon \cdot \frac{1}{6\epsilon} = \sigma_{ia}. \quad (4)$$

(Verified numerically: ratio 1.00000000.) So (1) is calibrated, and ϵ sets the interface width without changing its energy.

Two properties of h matter. $h(0) = 0$, $h(1) = 1$ give $f_w(0) = \sigma_{as}$ (wall against vapour) and $f_w(1) = \sigma_{is}$ (wall against ice). And $h'(0) = h'(1) = 0$ makes the wall term inert in both bulk phases: it acts only where the ice–vapour interface meets the wall, and cannot shift the bulk equilibria. A linear interpolant would violate the second property and bias ϕ all along Γ .

Note that (2) contains no σ_{ia} — only the two substrate energies. σ_{ia} appears in $\mathcal{F}_{\text{bulk}}$, and the ratio of the two is what produces $\cos \theta$ in §2.1.

1.1 Notation

Following Moure & Fu, the interface energies are σ_{ia} , σ_{is} and σ_{as} [J m^{-2}]. σ_{ia} is the only one the bulk has — the model is two-phase — and σ_{is}, σ_{as} enter solely through f_w .

Earlier revisions of this code also carried the surface-tension parameters $\Sigma_k = \sigma_{ka} + \sigma_{kb} - \sigma_{ab}$, i.e. $\Sigma_i = \sigma_{ia} + \sigma_{is} - \sigma_{as}$ and so on. **Those were removed on 2026-09-17.** They are a three-phase construction: each Σ_k weights the well of a bulk phase k , and here sediment is not a bulk phase. With a single ice–air interface the Σ_k all collapse to σ_{ia} , so carrying them meant two names for one number — which is precisely how the tree came to hold $\Sigma_i = 0.109$ (the two-phase σ_{ia}) alongside $\Sigma_a = 0.132$ (a stale three-phase value left from the abandoned sediment-phase model). The double-well coefficient is now simply σ_{ia} .

If sediment is ever reintroduced as a bulk phase field, the Σ_k belong with that model. They do not belong here.

2. First variation and the weak form

Vary (1) with an arbitrary test function η :

$$\delta\mathcal{F}[\eta] = \int_{\Omega} \left[6\sigma_{ia}\epsilon \nabla\phi \cdot \nabla\eta + \frac{6\sigma_{ia}}{\epsilon} f_1(\phi) \eta \right] d\Omega + \oint_{\Gamma} f'_w(\phi) \eta d\Gamma, \quad (5)$$

where

$$f_1(\phi) = \frac{1}{2} \frac{d}{d\phi} [\phi^2(1-\phi)^2] = \phi(1-\phi)(1-2\phi) \quad (6)$$

is the half-normalised double-well derivative the code uses — the normalisation that makes (3) have width exactly ϵ .

Equation (5) is the weak form: the volume integral is never integrated by parts in the Galerkin discretisation, and the boundary integral enters alongside it directly. That is the whole mechanism — the wall term is a surface integral in the residual, nothing more.

2.1 Strong form and the natural boundary condition

To see what boundary condition (5) encodes, integrate the gradient term by parts (for this purpose only):

$$\int_{\Omega} 6\sigma_{ia}\epsilon \nabla\phi \cdot \nabla\eta d\Omega = - \int_{\Omega} 6\sigma_{ia}\epsilon \nabla^2\phi \eta d\Omega + \oint_{\partial\Omega} 6\sigma_{ia}\epsilon (\partial_n\phi) \eta dS. \quad (7)$$

Substituting into (5) and collecting,

$$\delta\mathcal{F}[\eta] = \int_{\Omega} \left[-6\sigma_{ia}\epsilon \nabla^2\phi + \frac{6\sigma_{ia}}{\epsilon} f_1 \right] \eta d\Omega + \oint_{\partial\Omega} \left[6\sigma_{ia}\epsilon \partial_n\phi + f'_w(\phi) \right] \eta dS. \quad (8)$$

η is arbitrary in Ω and independently arbitrary on $\partial\Omega$, so both brackets vanish. The second is the natural boundary condition:

$$6\sigma_{ia}\epsilon \partial_n\phi + f'_w(\phi) = 0. \quad (9)$$

Where no wall term is declared, $f'_w \equiv 0$ and (9) reduces to $\partial_n\phi = 0$ — the zero-flux condition the solver had everywhere before this feature, which is a 90° contact angle imposed by omission.

With $f'_w = (\sigma_{is} - \sigma_{as})h'$ and $h'(\phi) = 6\phi(1-\phi)$,

$$\partial_n\phi = \frac{\sigma_{as} - \sigma_{is}}{6\sigma_{ia}\epsilon} h'(\phi) = \underbrace{\frac{\sigma_{as} - \sigma_{is}}{\sigma_{ia}}}_{\cos\theta} \frac{\phi(1-\phi)}{\epsilon}. \quad (10)$$

Young's equation was not assumed. The wall energy contributed only $\sigma_{as} - \sigma_{is}$; the σ_{ia} came from the bulk coefficient in (1). Their ratio is Young's $\cos \theta$, and it appears because the two terms of the same functional were each calibrated independently — the bulk to σ_{ia} , the wall to σ_{as} and σ_{is} .

By (3) the right-hand side of (10) is $\cos \theta |\nabla \phi|$, so writing $\nabla \phi = |\nabla \phi| \mathbf{m}$ with $\mathbf{m}$ the interface normal into the ice,

$$\boxed{\mathbf{m} \cdot \mathbf{n} = \cos \theta} \quad (11)$$

with $\mathbf{n}$ the outward wall normal: the contact angle measured through the ice. Limits: $\cos \theta = +1$ gives $\mathbf{m} = \mathbf{n}$, ice sheeted flat onto the wall, $\theta = 0$; $\cos \theta = -1$ gives $\theta = 180^\circ$, point contact.

3. The residual, and the mobility

The evolution is the gradient flow $\partial_t \phi = -L \delta \mathcal{F} / \delta \phi$, so with $\eta = N_a$ the residual is

$$R_a = \int_{\Omega} N_a \phi_t d\Omega + L \left\{ \int_{\Omega} \left[6\sigma_{ia} \epsilon \nabla N_a \cdot \nabla \phi + \frac{6\sigma_{ia}}{\epsilon} f_1 N_a \right] d\Omega + \oint_{\Gamma} f'_w(\phi) N_a d\Gamma \right\}. \quad (12)$$

The code does not store L . It stores the single number `mob_sub` = M , and matching (12) against the implemented interior form

$$R_a = \int N_a \phi_t + 3M\epsilon \int \nabla N_a \cdot \nabla \phi + \frac{3M}{\epsilon} \int f_1 N_a - (\text{source}) \quad (13)$$

shows what that number is:

$$L \cdot 6\sigma_{ia} = 3M \quad \implies \quad \boxed{M = 2\sigma_{ia}L} \quad (14)$$

One relation, fixed independently by both bulk coefficients — a consistency check, not a definition.

M is therefore not a free parameter: **it already contains the interface energy**, bundled with the kinetic mobility. This is not a choice made for the wall term; it is how `mob_sub` has always been computed. σ_{ia} reaches the solver through the capillary length $d_0 = \sigma_{ia} V_m / (RT)$, which sets $\tau_{\text{sub}} \propto 1/\sigma_{ia}$ and hence $M = \epsilon / (3\tau_{\text{sub}}) \propto \sigma_{ia}$. Verified directly: scaling `d0_sub0` over a factor of four scales `mob_sub` by the same factor, leaving $M/(2\sigma_{ia})$ constant to all printed digits.

3.1 The wall term, added naively

With $f'_w = (\sigma_{is} - \sigma_{as})h'$ and (14),

$$R_a^\Gamma = L(\sigma_{is} - \sigma_{as}) h'(\phi) N_a = \frac{M}{2\sigma_{ia}} (\sigma_{is} - \sigma_{as}) h'(\phi) N_a = \boxed{-3M \cos \theta \frac{h'(\phi)}{6} N_a} \quad (15)$$

and, the wall energy having no ϕ_t dependence, a Jacobian with no shift term,

$$J_{ab}^\Gamma = -3M \cos \theta \frac{h''(\phi)}{6} N_a N_b, \quad h''(\phi) = 6(1 - 2\phi). \quad (16)$$

Equation (15) is `assembly.c` verbatim: `Rb[a][0] = -rwb * 3.0 * mob_b * costhet * (dh/6.0) * N0b[a]`.

Nothing was rescaled and no prefactor was introduced. The σ_{ia} carried in by f'_w is the same σ_{ia} already inside M , so it divides back out; writing it explicitly would count it twice. The 3 and the 1/6 are not physics either — $L\sigma_{ia} = M/2 = 3M/6$, factored that way only so the 3 mirrors $3M\epsilon$ and $3M/\epsilon$ in (13).

3.2 Why no interface energy can survive

M is calibrated to deliver a requested kinetic coefficient β , not an energy scale. By (14) that forces $L \propto 1/\sigma_{ia}$, so no interface energy can appear in any residual coefficient. Had σ_{ia} remained in (15), that would have been the error. ϵ cancels for the same structural reason: neither f'_w nor L carries one.

4. Clamping, and why it is required

$h'(\phi) = 6\phi(1 - \phi)$ changes sign for $\phi < 0$ and $\phi > 1$. So does (15), and since $R = N\phi_t + \dots = 0$, a sign-flipped wall term drives ϕ *further* out of range: a runaway.

Nothing in the bulk opposes it. The interior form evaluates f_1 and the localiser at the clamped $\phi_c \in [0, 1]$, so for $\phi < 0$ both are identically zero — there is no restoring force outside the range by construction. An unclamped wall term is then the only term acting there, and it acts outward.

This is not hypothetical. At $\epsilon/R = 1/100$ a -2×10^{-6} excursion at the wall grew to -0.05 within a few hundred steps, crossed `-phase_lo`, and the residual's domain guard then returned a zero residual, which SNES read as $\|F\| = 0$ and reported as converged. The solution froze for the remaining 23,370 steps while the clock ran on to t_{final} and the run reported success. The implementation therefore clamps ϕ to $[0, 1]$ before evaluating h' , with the Jacobian clamping identically and carrying the chain-rule factor $d\phi_c/d\phi$, so that it remains an exact Jacobian of the clamped residual.

5. Using it

<code>-wall_faces "y0,y1"</code>	which domain faces are regolith
<code>-sigma_is</code>	ice-regolith interface energy [J m^{-2}]
<code>-sigma_as</code>	air-regolith interface energy [J m^{-2}]
<code>-sigma_ia</code>	ice-air interface energy; default <code>-sigma_ia_bulk</code>

The solver forms $\cos \theta = (\sigma_{as} - \sigma_{is})/\sigma_{ia}$, refuses the run if $|\sigma_{as} - \sigma_{is}| > \sigma_{ia}$ (no angle exists; one phase wets completely), and prints both the energies and the derived θ .

σ_{ia} defaults to `-sigma_ia_bulk`: the wall sees the same ice-air interface the bulk does, so by default there is one number rather than two. They are separable only so a sensitivity test can perturb the wall's view of it independently.

The only admissibility constraint is $|\sigma_{as} - \sigma_{is}| \leq \sigma_{ia}$. The three-phase triple-well constraint $\sigma_{is} > \frac{\sigma_{ia}}{2}(1 - \cos \theta)$ does *not* apply: it came from requiring $\Sigma_s > 0$, and there is no sediment phase and no triple well here.

6. Validation

Five prescribed angles from 30° to 150° in a flat channel recover Young's prediction with a maximum error of 0.127° and an RMS of 0.062° , against a 2° acceptance criterion, with the error falling as ϵ/R is refined. A droplet on a single wall, which has no confinement, agrees with the channel, so the angle is set by the boundary condition rather than by the geometry. Details in `studies/contact_angle/`.